GitOps doesn't enforce specific technologies, but it's best implemented with Git and Kubernetes. That will be our focus.

The GitOps Working Group, part of the CNCF, defines GitOps in terms of four principles (https://opengitops.dev):

1. *Declarative*—"A system managed by GitOps must have its desired state expressed declaratively."
    - Working with Kubernetes, we can express the desired state via YAML files (manifests).
    - Kubernetes manifests declare what we want to achieve, not how. The platform is responsible for finding a way to achieve the desired state.
2. *Versioned and immutable*—"Desired state is stored in a way that enforces immutability, versioning and retains a complete version history."
    - Git is the preferred choice for ensuring the desired state is versioned and the whole history retained. That makes it possible, among other things, to roll back to a previous state with ease.
    - The desired state stored in Git is immutable and represents the single source of truth.